

# Residence time function and demo

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## Overview

This is a simple R simulation of some arbitrary pool with discrete units flowing in and out. It is meant to explore the concept of residence time.

## Packages

```
library(data.table)
```

## The simulation function

```
rtsim <- function(p0, r_in, r_out, dt, time) {  
  
  # Create initial pool  
  p <- data.table(id = 1:p0, age = 0)  
  
  n <- time %/% dt  
  
  # Loop through times  
  for (t in 1:n) {  
  
    # Add new ids  
    new_ids <- 1:(r_in * dt) + max(p$id)  
  
    # Select random ids for removal  
    old_ids <- sample(p$id, r_out * dt)  
  
    # Update age  
    p$age <- p$age + dt  
  
    # Update pool  
    p <- rbind(p[!id %in% old_ids, ], data.table(id = new_ids, age = 0))  
  
  }  
  
  return(p)  
}
```

# Application

1. Pool of 10,000 with flow rate in = flow rate out = 100.

```
out <- rtsim(1E4, 100, 100, 1, 1000)
mean(out$age)
```

```
## [1] 98.6747
```

2. Increase flow.

```
out <- rtsim(1E4, 200, 200, 1, 1000)
mean(out$age)
```

```
## [1] 48.3969
```

3. Decrease flow

```
out <- rtsim(1E4, 10, 10, 1, 1000)
mean(out$age)
```

```
## [1] 631.0878
```

4. Flow in > flow out (pool size grows)

```
out <- rtsim(1E4, 102, 100, 1, 1000)
mean(out$age)
```

```
## [1] 114.5123
```

```
nrow(out)
```

```
## [1] 12000
```

```
nrow(out) / 102
```

```
## [1] 117.6471
```

```
nrow(out) / 100
```

```
## [1] 120
```

5. Flow out > flow in

```
out <- rtsim(1E4, 100, 102, 1, 1000)
mean(out$age)
```

```
## [1] 80.69
```

```
nrow(out) / 102
```

```
## [1] 78.43137
```

```
nrow(out) / 100
```

```
## [1] 80
```